

§5.3

The Binomial Distribution

Counting the number of successes in a fixed number of trials

Today's Goal

After completing this section, you should be able to:

- Identify situations that can be modeled by a binomial distribution
- Use the binomial probability formula to calculate probabilities
- Use Excel to calculate binomial probabilities
- Find the mean and standard deviation of a binomial distribution
- Determine whether an observed result is significantly high or low

Learning Target 8

LT.8 Discrete Probability Distributions

I understand what a discrete probability distribution is (including binomial distributions). I can represent the discrete probability distribution of a given random variable by creating an appropriate table or graph and can use it to calculate the probability of an event. I can also calculate the mean, variance, and standard deviation for a discrete probability distribution.

Binomial Experiments

Definition 5.3.1

A **binomial experiment** is a probability experiment that satisfies the following requirements:

1. The experiment has a **fixed number of trials**, denoted n .
2. Every outcome can be classified as either a **success** (S) or **failure** (F).
3. The trials are **independent**—the outcome of one trial does not affect the outcomes of other trials.
4. The **probability of success** is the same for each trial, denoted p .

Notation for a Binomial Experiment

The random variable X for a binomial experiment counts the number of successes in the n trials.

For a binomial experiment, we use the following notation:

n = the fixed number of trials

p = the probability of success on one trial

$q = 1 - p$ = the probability of failure on one trial

X = the number of successes among the n trials

The notation $X \sim B(n, p)$ means “ X is a random variable for a binomial distribution with n trials and probability of success p .”

Is This a Binomial Experiment?

A fair 4-sided die is rolled 20 times. Let X represent the number of times a 3 is rolled. Is this a binomial experiment? If so, identify n , p , and q .

We check the four requirements:

1. **Fixed number of trials:** Yes, we roll the die 20 times, so $n = 20$.
2. **Two outcomes:** Yes, each roll is either a “success” (rolling a 3) or a “failure” (rolling anything else).
3. **Independent trials:** Yes, the outcome of each roll does not affect other rolls.
4. **Constant probability:** Yes, $P(\text{rolling a 3}) = \frac{1}{4}$ for every roll.

Identify n , p , and q

A fair 4-sided die is rolled 20 times. Let X represent the number of times a 3 is rolled.

This is a binomial experiment with

$$n = 20$$

$$p = \frac{1}{4}$$

$$q = \frac{3}{4}.$$

We can write $X \sim B\left(20, \frac{1}{4}\right)$.

Now You Try: Binomial Experiments

You Try 5.3.1. A basketball player makes 85% of her free throws. Each free throw is either made or missed.

At practice, she shoots free throws one after another until her first miss, and counts how many she shot. Let X be the number of free throws she shoots before she stops.

- (a) Is this a binomial experiment?
- (b) If it is, identify n , p and q .

You Try 5.3.1: Checking Your Answer

(a) Is this a binomial experiment?

No — the number of trials is not fixed in advance.

(b) If it is, identify n , p and q .

There is no n to identify — how many trials there are is decided by how they turn out rather than settled beforehand.

p and q are a different matter: the chance that she makes a free throw is stated, and it is the same every time, so both are perfectly well defined here.

Sampling Without Replacement

When sampling without replacement, the trials are technically dependent because removing one item changes the probabilities for subsequent selections. However, if the sample is small relative to the population, this dependence is negligible.

5% GUIDELINE

When sampling **without replacement**, if the sample size is no more than 5% of the population size, then we may treat the selections as independent events. This allows us to use the binomial distribution as an approximation.

The Binomial Probability Formula

Theorem 5.3.3 Binomial Probability Formula

For a binomial experiment, the probability of getting exactly x successes in n trials is:

$$P(X = x) = \frac{n!}{(n - x)! \cdot x!} \cdot p^x \cdot q^{n-x}$$

where $n!$ (“ n factorial”) is the product of all positive integers up to n :

$$n! = n \cdot (n - 1) \cdot (n - 2) \cdots 3 \cdot 2 \cdot 1$$

By definition, $0! = 1$.

Exactly 4 Heads in 6 Flips

A weighted coin comes up heads 70% of the time. Calculate the probability of getting exactly 4 heads when the coin is flipped 6 times. Round to 4 decimal places.

What are n , p , q , and x ?

We have $n = 6$, $p = 0.70$, $q = 0.30$, and $x = 4$.

Exactly 4 Heads in 6 Flips

We have $n = 6$, $p = 0.70$, $q = 0.30$, and $x = 4$. Using the formula:

$$\begin{aligned}P(X = 4) &= \frac{6!}{(6 - 4)! \cdot 4!} \cdot (0.70)^4 \cdot (0.30)^{6-4} \\&= \frac{6!}{2! \cdot 4!} \cdot (0.70)^4 \cdot (0.30)^2 \\&= \frac{720}{(2)(24)} \cdot (0.2401) \cdot (0.09) \\&= 15 \cdot 0.2401 \cdot 0.09 \\&\approx 0.3241.\end{aligned}$$

There is about a 32.41% chance of getting exactly 4 heads.

Using Excel for Binomial Probabilities

Procedure 5.3.5 Using Excel for Binomial Probabilities

The Excel function `BINOM.DIST` calculates binomial probabilities:

`=BINOM.DIST(x, n, p, cumulative)`

where

- x = number of successes;
- n = number of trials;
- p = probability of success (expressed as a decimal number);
- `cumulative`:
 - `FALSE` calculates $P(X = x)$ (exactly x successes)
 - `TRUE` calculates $P(X \leq x)$ (at most x successes)

Using Excel for Binomial Probabilities

Procedure 5.3.5, continued

Examples:

- $P(X = 3)$ with $n = 5, p = 0.85$: `=BINOM.DIST(3,5,0.85,FALSE)`
- $P(X \leq 12)$ with $n = 20, p = 0.41$:
`=BINOM.DIST(12,20,0.41,TRUE)`

At Most 14 of 25 Students

Suppose 62% of the students at a large university live off campus. If 25 students are randomly selected, find the probability that at most 14 of them live off campus. Round to 4 decimal places.

We have $X \sim B(25, 0.62)$ and want to find $P(X \leq 14)$.

This cumulative probability would require calculating

$$P(X \leq 14) = P(X = 0) + P(X = 1) + P(X = 2) + \cdots + P(X = 14).$$

Instead, we use Excel: `=BINOM.DIST(14, 25, 0.62, TRUE)` returns 0.3355.

There is about a 33.55% chance that at most 14 of the 25 students live off campus.

Now You Try: Binomial Probabilities

You Try 5.3.2. At a coffee shop, 35% of customers order a pastry. Let X be the number of customers who order a pastry in a random sample of 12 customers.

Use Excel's `=BINOM.DIST ( )` function. Round each probability to **four** decimal places.

- (a) Find the probability that exactly 4 of the 12 customers order a pastry.
- (b) Find the probability that at least 6 of the 12 customers order a pastry.

You Try 5.3.2: Checking Your Answer

(a) Exactly 4 of the 12 customers

=BINOM.DIST(4, 12, 0.35, FALSE) returns $P(X = 4) \approx 0.2367$.

(b) At least 6 of the 12 customers

“At least 6” lets in 6 itself and every count above it.

=1-BINOM.DIST(5, 12, 0.35, TRUE) returns

$P(X \geq 6) = 1 - P(X \leq 5) \approx 0.2127$.

Parameters of a Binomial Distribution

For binomial distributions, there are simple formulas for the mean and standard deviation.

Theorem 5.3.7 Parameters of a Binomial Distribution

For a binomial distribution with n trials and probability of success p ,

$$\text{Mean: } \mu = np$$

$$\text{Variance: } \sigma^2 = npq$$

$$\text{Standard Deviation: } \sigma = \sqrt{npq}.$$

Mean and Standard Deviation

For the situation in Example 3 (25 students, 62% living off campus), find the mean and standard deviation. Round to 2 decimal places.

We have $n = 25$, $p = 0.62$, and $q = 0.38$.

$$\mu = np = (25)(0.62) = 15.50$$

$$\sigma = \sqrt{npq} = \sqrt{(25)(0.62)(0.38)} = \sqrt{5.89} \approx 2.43.$$

On average, we expect 15.5 of the 25 students to live off campus, with a standard deviation of about 2.43 students.

Now You Try: Binomial Parameters

You Try 5.3.3. A streaming service finds that 60% of its users watch on a phone. Let X be the number of users who watch on a phone in a random sample of 18 users.

- (a) Find the mean of X . Round to **two** decimal places.
- (b) Find the standard deviation of X . Round to **two** decimal places.
- (c) For a random sample of 72 users, how do the mean and the standard deviation compare with the values you found above?

You Try 5.3.3: Checking Your Answer

(a) Mean

$$\mu = np = (18)(0.60) = 10.80$$

(b) Standard deviation

$$\sigma = \sqrt{npq} = \sqrt{(18)(0.60)(0.40)} = \sqrt{4.32} \approx 2.08$$

(c) A random sample of 72 users

$$\mu = (72)(0.60) = 43.20$$

$$\sigma = \sqrt{(72)(0.60)(0.40)} = \sqrt{17.28} \approx 4.16$$

The mean is 4 times as large, and the standard deviation is larger but less than 4 times as large.

Identifying Significant Results

Definition 5.3.9

For a probability distribution with mean μ and standard deviation σ , an observed value x is a **significantly low** result if it lies more than two standard deviations below the mean:

$$x < \mu - 2\sigma.$$

An observed value is a **significantly high** result if it lies more than two standard deviations above the mean:

$$x > \mu + 2\sigma.$$

Significance Thresholds

Definition 5.3.9, continued

Values between $\mu - 2\sigma$ and $\mu + 2\sigma$ are considered **not significant**—they fall within the range of typical outcomes.

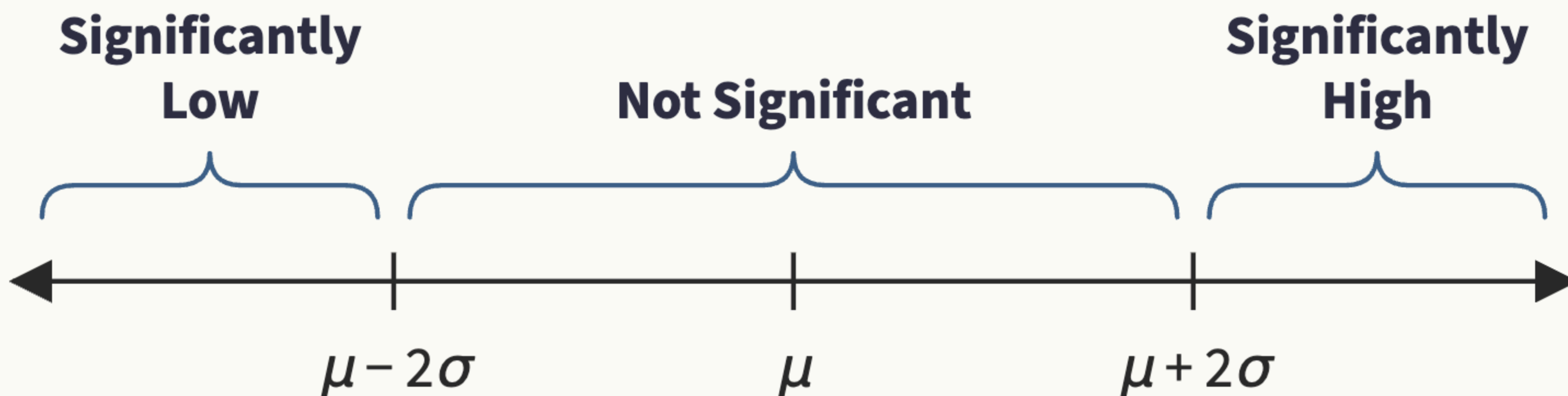


Figure 5.3.10. Significance thresholds.

Is 252 Wins Significantly High?

Between 1974 and 2011, there were 460 NFL football games decided in overtime, and 252 of them were won by the team that won the overtime coin toss. Is the result of 252 wins significantly high? Assume winning and losing are equally likely after winning the coin toss.

Let X represent the number of games won by the team that won the coin toss. If winning the coin toss provides no advantage, then $p = 0.5$. So we have $X \sim B(460, 0.5)$.

Is 252 Wins Significantly High?

Calculate the mean and standard deviation:

$$\mu = np = (460)(0.5) = 230$$

$$\sigma = \sqrt{npq} = \sqrt{(460)(0.5)(0.5)} = \sqrt{115} \approx 10.72.$$

The threshold for “significantly high” is

$$\mu + 2\sigma = 230 + 2(10.72) = 230 + 21.44 = 251.44.$$

Since $252 > 251.44$, the result of 252 wins is **significantly high** (though just barely). This suggests that winning the overtime coin toss may provide a real advantage.

A Genetics Experiment

A genetics experiment has a 25% probability of producing a certain trait. If 60 offspring are observed:

(a) Find the mean and standard deviation.

$$\mu = (60)(0.25) = 15 \text{ and } \sigma = \sqrt{(60)(0.25)(0.75)} = \sqrt{11.25} \approx 3.35$$

(b) How many offspring with the trait would be significantly high?

$$\mu + 2\sigma = 15 + 2(3.35) = 21.70, \text{ so } 22 \text{ or more offspring.}$$

(c) If 22 offspring have the trait, is this unusual?

Yes. Since $22 > 21.70$, the result is significantly high.